

WPA2 Client Connection Analysis Using Wireshark

1. Objective

The objective of this exercise is to understand the process followed by a wireless client while connecting to an access point configured with WPA2 security. The captured packets are analysed using Wireshark to study the 802.11 management frame exchange, security-related Information Elements, WPA2 4-way handshake and available radio information.

Capture Note: Due to limitations in capturing complete raw 802.11 frames using the available personal laptop and wireless adapter, a sample 802.11/WPA capture was used for packet-level learning and analysis.

2. WPA2 Client Connection Overview

The connection process begins with wireless network discovery, followed by 802.11 authentication and association. After association, the WPA2 4-way handshake establishes the encryption keys required for protected communication.

4. Packet Analysis

4.1 Probe Request

Purpose:

Client searches for the required wireless network.

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Cisco_70:18:d0	Broadcast	802.11	298	Beacon frame, SN=3039, FN=0, Flags=....., BI=102, SSID="ikeriri-5g"
2	37.245000	Sony_50:73:db	Broadcast	802.11	130	Probe Request, SN=379, FN=0, Flags=....., SSID=Wildcard (Broadcast)
3	37.247000	Cisco_70:18:d0	Sony_50:73:db	802.11	292	Probe Response, SN=1748, FN=0, Flags=....R..., BI=102, SSID="ikeriri-5g"
4	50.744000	Sony_50:73:db	Cisco_70:18:d0	802.11	54	Authentication, SN=482, FN=0, Flags=.....
5	50.744000	Cisco_70:18:d0	Sony_50:73:db	802.11	54	Authentication, SN=3802, FN=0, Flags=.....
6	50.744000	Sony_50:73:db	Cisco_70:18:d0	802.11	243	Association Request, SN=483, FN=0, Flags=....., SSID="ikeriri-5g"
7	50.746000	Cisco_70:18:d0	Sony_50:73:db	802.11	173	Association Response, SN=3803, FN=0, Flags=.....
8	50.746000	Sony_50:73:db	Cisco_70:18:d0	EAPOL	179	Key (Message 1 of 4)
9	50.789000	Sony_50:73:db	Cisco_70:18:d0	EAPOL	179	Key (Message 2 of 4)
10	50.798000	Cisco_70:18:d0	Sony_50:73:db	EAPOL	213	Key (Message 3 of 4)
11	50.798000	Sony_50:73:db	Cisco_70:18:d0	EAPOL	157	Key (Message 4 of 4)
12	50.799000	Cisco_9c:6a:e4	Sony_50:73:db	802.11	132	QoS Data, SN=0, FN=0, Flags=p....F.
13	50.990000	Sony_50:73:db	Broadcast	802.11	408	QoS Data, SN=0, FN=0, Flags=p....T
14	50.990000	Modacom_94:ea:bc	Sony_50:73:db	802.11	662	QoS Data, SN=1, FN=0, Flags=p....F.
15	51.126000	Sony_50:73:db	Broadcast	802.11	102	QoS Data, SN=1, FN=0, Flags=p....T
16	92.162000	Sony_50:73:db	Cisco_70:18:d0	802.11	50	Disassociate, SN=966, FN=0, Flags=.....

Frame 2: Packet, 130 bytes on wire (1040 bits), 130 bytes captured (1040 bits)

- Radiotap Header v0, Length 24
- 802.11 radio information
- IEEE 802.11 Probe Request, Flags:
- IEEE 802.11 Wireless Management
 - Tagged parameters (82 bytes)
 - Tag: SSID parameter set: Wildcard SSID
 - Tag: Supported Rates 6, 9, 12, 18, 24, 36, 48, 54, [Mbit/sec]
 - Tag: DS Parameter set: Current Channel: 36
 - Tag: HT Capabilities
 - Tag: Vendor Specific: Microsoft Corp.: Unknown 8
 - Tag: VHT Capabilities
 - Tag: Extended Capabilities (4 octets)
 - Tag: Interworking
 - Tag: Vendor Specific: Wi-Fi Alliance: Hotspot 2.0 Indication

0000 00 00 18 00 6f 00 00 00 7c e1 ec 60 60 ae 0a 00o...|...
0010 00 0c 3c 14 40 01 ce a3 40 00 00 00 ff ff ff ff ..<@...@...
0020 ff ff 40 40 a7 50 73 db ff ff ff ff ff b0 17 ...Ps...
0030 00 00 01 08 0c 12 18 24 30 48 60 6c 03 01 24 2d\$0H'l\$-
0040 1a 6e 01 03 ff 00 00 00 00 00 00 00 00 00 00 ..n.....
0050 00 00 00 00 00 00 00 00 00 00 dd 07 00 50 f2P...
0060 08 00 00 00 bf 0c 20 51 80 31 fc ff 00 00 fc ffQ_1.....
0070 00 00 7f 04 00 00 00 80 6b 01 0f dd 05 50 6f 9ak...Po
0080 10 00 ..

4.2 Probe Response

Purpose:

AP responds with network and capability information.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Cisco_70:18:d0	Broadcast	802.11	298	Beacon frame, SN=3039, FN=0, Flags=....., BI=102, SSID="ikeriri-5g"
2	37.245000	Sony_50:73:db	Broadcast	802.11	130	Probe Request, SN=379, FN=0, Flags=....., SSID=Wildcard (Broadcast)
3	37.247000	Cisco_70:18:d0	Sony_50:73:db	802.11	292	Probe Response, SN=1748, FN=0, Flags=....R..., BI=102, SSID="ikeriri-5g"
4	50.744000	Sony_50:73:db	Cisco_70:18:d0	802.11	54	Authentication, SN=182, FN=0, Flags=...

IEEE 802.11 Probe Response, Flags:R... IEEE 802.11 Wireless Management Fixed parameters (12 bytes) Timestamp: 322324815363 Beacon Interval: 0.104448 [Seconds] Capabilities Information: 0x0111 Tagged parameters (232 bytes) Tag: SSID parameter set: "ikeriri-5g" Tag: Supported Rates 6(B), 9(B), 12(B), 18(B), 24(B), 36(B), 48(B), 54(B), [Mbps] Tag: HT Capabilities Tag: RSN Information Tag Number: RSN Information (48) Tag length: 20 RSN Version: 1 Group Cipher Suite: 00:0f:ac (Ieee 802.11) AES (CCM) Pairwise Cipher Suite Count: 1 Pairwise Cipher Suite List 00:0f:ac (Ieee 802.11) AES (CCM) Auth Key Management (AKM) Suite Count: 1 Auth Key Management (AKM) List 00:0f:ac (Ieee 802.11) PSK RSN Capabilities: 0x003c Tag: HT Operation Tag: Extended Capabilities (8 octets) Tag: Cisco CCX1 CKIP + Device Name Tag: VHT Capabilities Tag: VHT Operation Tag: Tx Power Envelope Tag: Vendor Specific: Microsoft Corp.: WMM/WME: Parameter Element Tag: Vendor Specific: Cisco Systems, Inc: Aironet Unknown (1) (1) Tag: Vendor Specific: Cisco Systems, Inc: Aironet CCX version = 5 Tag: Vendor Specific: Cisco Systems, Inc: Aironet Unknown (11) (11) Tag: Vendor Specific: Cisco Systems, Inc: Aironet Client MFP Enabled Tag: Vendor Specific: Cisco Systems, Inc: Aironet Unknown (24)	0000 00 00 18 00 6f 00 00 00 7c e1 ec 60 30 b6 0a 00 0010 00 0c 3c 14 40 01 d4 a3 50 08 3c 00 40 40 a7 50 0020 73 db 50 0f 80 70 18 d0 50 0f 80 70 18 d0 40 6d 0030 03 5e 0e 0c 4b 00 00 00 66 00 11 01 00 0a 69 6b 0040 65 72 69 72 69 2d 35 67 01 08 8c 92 98 a4 b0 c8 0050 e0 ec 2d 1a ee 19 1b ff ff 00 00 00 00 00 00 00 0060 00 00 00 00 00 00 00 00 00 00 00 00 00 30 14 0070 01 00 00 0f ac 04 01 00 00 0f ac 04 01 00 00 0f 0080 ac 02 3c 00 3d 16 24 05 05 00 00 00 00 00 00 00 0090 00 00 00 00 00 00 00 00 00 00 00 00 7f 08 00 00 00a0 00 00 00 00 00 40 85 1a 00 00 a5 00 0f 00 ff 03 00b0 19 00 61 70 00 00 00 00 00 00 00 00 00 00 00 00c0 00 00 05 00 00 53 bf 0c b2 59 82 0f fa ff 00 00 00d0 fa ff 00 00 c0 05 01 2a 00 c0 ff c3 04 02 02 02 00e0 02 dd 18 00 50 f2 02 01 01 82 00 03 a4 00 00 27 00f0 a4 00 00 42 43 5e 00 62 32 2f 00 dd 06 00 40 96 0100 01 01 00 dd 05 00 40 96 03 05 dd 05 00 40 96 0b 0110 09 dd 05 00 40 96 14 01 dd 0a 00 40 96 18 ac 04 0120 01 00 00 0f
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4.3 Information Elements and RSN

Purpose:

To examine the network capabilities and WPA2 security information advertised by the AP.

Look for:

- SSID
- Supported Rates
- Channel-related information
- RSN Information

4.4 Authentication

Important observation:

This management-frame authentication should not be confused with the WPA2 4-way handshake.

```
▶ Frame 4: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits)
▶ Radiotap Header v0, Length 24
▼ 802.11 radio information
  PHY type: 802.11a (OFDM) (5)
  Turbo type: Non-turbo (0)
  Data rate: 6.0 Mb/s
  Channel: 36
  Frequency: 5180MHz
  Signal strength (dBm): -64 dBm
  Noise level (dBm): -95 dBm
  Signal/noise ratio (dB): 31 dB
  TSF timestamp: 854700118040970
  ▶ [Duration: 64µs]
▼ IEEE 802.11 Authentication, Flags: .....
  Type/Subtype: Authentication (0x000b)
  ▶ Frame Control Field: 0xb000
    .000 0000 0011 1100 = Duration: 60 microseconds
  ▶ Receiver address: Cisco_70:18:d0 (50:0f:80:70:18:d0)
  ▶ Destination address: Cisco_70:18:d0 (50:0f:80:70:18:d0)
  ▶ Transmitter address: Sony_50:73:db (40:40:a7:50:73:db)
  ▶ Source address: Sony_50:73:db (40:40:a7:50:73:db)
  ▶ BSS Id: Cisco_70:18:d0 (50:0f:80:70:18:d0)
    .... .... 0000 = Fragment number: 0
    0001 1110 0010 .... = Sequence number: 482
    [WLAN Flags: .....]
▼ IEEE 802.11 Wireless Management
  ▼ Fixed parameters (6 bytes)
    Authentication Algorithm: Open System (0)
    Authentication SEQ: 0x0001
    Status code: Successful (0x0000)
```

4.5 Association Request

The client requests to join the AP and advertises its capabilities.

```
▶ Frame 6: Packet, 243 bytes on wire (1944 bits), 243 bytes captured (1944 bits)
▶ Radiotap Header v0, Length 24
▶ 802.11 radio information
▶ IEEE 802.11 Association Request, Flags: .....
▼ IEEE 802.11 Wireless Management
  ▶ Fixed parameters (4 bytes)
  ▼ Tagged parameters (191 bytes)
    ▶ Tag: SSID parameter set: "ikeriri-5g"
    ▶ Tag: Supported Rates 6(B), 9(B), 12(B), 18(B), 24(B), 36(B), 48(B), 54(B), [Mbit/sec]
    ▶ Tag: Power Capability Min: 13, Max: 23
    ▶ Tag: Supported Channels
    ▼ Tag: RSN Information
      Tag Number: RSN Information (48)
      Tag length: 20
      RSN Version: 1
      ▶ Group Cipher Suite: 00:0f:ac (Ieee 802.11) AES (CCM)
      Pairwise Cipher Suite Count: 1
      ▶ Pairwise Cipher Suite List 00:0f:ac (Ieee 802.11) AES (CCM)
      Auth Key Management (AKM) Suite Count: 1
      ▶ Auth Key Management (AKM) List 00:0f:ac (Ieee 802.11) PSK
      ▶ RSN Capabilities: 0x003c
    ▶ Tag: HT Capabilities
    ▶ Tag: Vendor Specific: Microsoft Corp.: WMM/WME: Information Element
    ▶ Tag: VHT Capabilities
    ▶ Tag: Extended Capabilities (8 octets)
    ▶ Tag: Extended Capabilities (4 octets)
```

4.6 Association Response

The AP accepts or rejects the association request.

```
▶ Frame 7: Packet, 173 bytes on wire (1384 bits), 173 bytes captured (1384 bits)
▶ Radiotap Header v0, Length 24
▶ 802.11 radio information
▶ IEEE 802.11 Association Response, Flags: .....
▼ IEEE 802.11 Wireless Management
  ▶ Fixed parameters (6 bytes)
  ▼ Tagged parameters (119 bytes)
    ▶ Tag: Supported Rates 6(B), 9(B), 12(B), 18(B), 24(B), 36(B), 48(B), 54(B), [Mbit/sec]
    ▶ Tag: Extended Capabilities (8 octets)
    ▶ Tag: HT Capabilities
    ▶ Tag: HT Operation
    ▶ Tag: VHT Capabilities
    ▶ Tag: VHT Operation
    ▶ Tag: Vendor Specific: Microsoft Corp.: WMM/WME: Parameter Element
```

5. WPA2 4-Way Handshake

Message 1 — AP to Client

Observation:

AP initiates the WPA2 key establishment process and sends its nonce-related information.

```
▶ Frame 8: Packet, 179 bytes on wire (1432 bits), 179 bytes captured (1432 bits)
▶ Radiotap Header v0, Length 24
▼ 802.11 radio information
  PHY type: 802.11a (OFDM) (5)
  Turbo type: Non-turbo (0)
  Data rate: 6.0 Mb/s
  Channel: 36
  Frequency: 5180MHz
  Signal strength (dBm): -44 dBm
  Noise level (dBm): -95 dBm
  Signal/noise ratio (dB): 51 dB
  TSF timestamp: 863290052632970
  ▶ [Duration: 232µs]
▶ IEEE 802.11 QoS Data, Flags: .....F.
▶ Logical-Link Control
▼ 802.1X Authentication
  Version: 802.1X-2004 (2)
  Type: Key (3)
  Length: 117
  Key Descriptor Type: EAPOL RSN Key (2)
  [Message number: 1]
  ▶ Key Information: 0x008a
  Key Length: 16
  Replay Counter: 1
  WPA Key Nonce: 15adf473164f43a34f211ebc34495b588af5b915c0dd4478f5fbc89d2f7bd0fa
  Key IV: 00000000000000000000000000000000
  WPA Key RSC: 0000000000000000
  WPA Key ID: 0000000000000000
  WPA Key MIC: 00000000000000000000000000000000
  WPA Key Data Length: 22
  ▶ WPA Key Data: dd14000fac04b9c9f71f0c96f62b6c11f545d2dff41b
```

Message 2 — Client to AP

Observation:

Client responds with its nonce-related information and key confirmation data.

```
▶ Frame 9: Packet, 179 bytes on wire (1432 bits), 179 bytes captured (1432 bits)
▶ Radiotap Header v0, Length 24
▼ 802.11 radio information
  PHY type: 802.11a (OFDM) (5)
  Turbo type: Non-turbo (0)
  Data rate: 6.0 Mb/s
  Channel: 36
  Frequency: 5180MHz
  Signal strength (dBm): -52 dBm
  Noise level (dBm): -95 dBm
  Signal/noise ratio (dB): 43 dB
  TSF timestamp: 1047973646360970
  ▶ [Duration: 232µs]
▶ IEEE 802.11 QoS Data, Flags: .....T
▶ Logical-Link Control
▼ 802.1X Authentication
  Version: 802.1X-2001 (1)
  Type: Key (3)
  Length: 117
  Key Descriptor Type: EAPOL RSN Key (2)
  [Message number: 2]
  ▶ Key Information: 0x010a
    Key Length: 0
    Replay Counter: 1
    WPA Key Nonce: 1b9717293f9d9d6979d94b36dbc9d83418bbce09f72edc1e1ae4fd79821ffda4
    Key IV: 00000000000000000000000000000000
    WPA Key RSC: 0000000000000000
    WPA Key ID: 0000000000000000
    WPA Key MIC: 2f8e7921e572afd75a7c898e625ffb43
    WPA Key Data Length: 22
  ▶ WPA Key Data: 3014010000fac040100000fac040100000fac023c00
```

Message 3 — AP to Client

Observation:

AP sends the information required for the next stage of key installation and group-key configuration.

```
▶ Frame 10: Packet, 213 bytes on wire (1704 bits), 213 bytes captured (1704 bits) on 0
▶ Radiotap Header v0, Length 24
▶ 802.11 radio information
  ▶ PHY type: 802.11a (OFDM) (5)
  ▶ Turbo type: Non-turbo (0)
  ▶ Data rate: 6.0 Mb/s
  ▶ Channel: 36
  ▶ Frequency: 5180MHz
  ▶ Signal strength (dBm): -44 dBm
  ▶ Noise level (dBm): -95 dBm
  ▶ Signal/noise ratio (dB): 51 dB
  ▶ TSF timestamp: 1086628352024970
  ▶ [Duration: 276µs]
▶ IEEE 802.11 QoS Data, Flags: .....F.
▶ Logical-Link Control
▶ 802.1X Authentication
  ▶ Version: 802.1X-2004 (2)
  ▶ Type: Key (3)
  ▶ Length: 151
  ▶ Key Descriptor Type: EAPOL RSN Key (2)
  ▶ [Message number: 3]
  ▶ Key Information: 0x13ca
  ▶ Key Length: 16
  ▶ Replay Counter: 2
  ▶ WPA Key Nonce: 15adf473164f43a34f211ebc34495b588af5b915c0dd4478f5fbc89d2f7bd0fa
  ▶ Key IV: 00000000000000000000000000000000
  ▶ WPA Key RSC: 0000000000000000
  ▶ WPA Key ID: 0000000000000000
  ▶ WPA Key MIC: e481fe9d4a2e0a53dd1119fb36104330
  ▶ WPA Key Data Length: 56
  ▶ WPA Key Data: b43c7737faedbf17306b5ea1d5059fd5379d0145fa6ddac1e08d460de93cd72c103a7fb2c21445d2b740139c80ac569ae1e865a198c79a21
```

Message 4 — Client to AP

Observation:

Client confirms successful key installation, completing the 4-way handshake.

[illegible]

6. Protected Data

After the successful WPA2 key establishment process, data communication takes place using protected wireless frames. The payload is encrypted using the negotiated security keys and cannot be directly interpreted without the appropriate decryption information.

```
▶ Frame 13: Packet, 408 bytes on wire (3264 bits), 408 bytes captured (3264 bits)
▶ Radiotap Header v0, Length 24
▼ 802.11 radio information
    PHY type: 802.11a (OFDM) (5)
    Turbo type: Non-turbo (0)
    Data rate: 9.0 Mb/s
    Channel: 36
    Frequency: 5180MHz
    Signal strength (dBm): -64 dBm
    Noise level (dBm): -96 dBm
    Signal/noise ratio (dB): 32 dB
    TSF timestamp: 1911262072856970
    ▶ [Duration: 364µs]
▶ IEEE 802.11 QoS Data, Flags: .p.....T
▼ Data (350 bytes)
    Data [...]: 28a5b2ed1557581d200a83231c14f1f2483acbb1719439d76beaab6d1a0243638f24023ec5d599d23
    [Length: 350]
```

6.Conclusion

This exercise provided practical understanding of the WPA2 client connection process through Wireshark packet analysis. The capture was used to identify the major stages of wireless network discovery, authentication, association and WPA2 key establishment. Analysis of the RSN Information Element helped identify the advertised security configuration, while the EAPOL frames demonstrated the 4-way handshake used to establish the keys required for protected communication.